

INTRODUCTION TO MACHINE LEARNING

Introduction and Logistics

Why build intelligent machines, a brief history of AI, what this course is about, and how it works as an experience.

A Bit About Me

- Joseph Bakarji · jb50@aub.edu.lb · josephbakarji.com
- Research at the intersection of **machine learning**, **scientific modeling**, and **music**
- Office hours and how to reach the teaching team: intro2ml.com, kept current there

Today's Plan

- Motivation: why you should (or should not) take this course
- A brief history of artificial intelligence
- What this course is about
- This course as a journey
- Logistics and course policy

PART I

Motivation

give me a sarcastic introductory paragraph for an introductory course on machine learning to share with students. include a motivation for why they should take the course and why it matters



Welcome to *Introduction to Machine Learning*—the course where we bravely attempt to teach you how to “make machines intelligent,” while most of your time will actually be spent making peace with matrices, probabilities, and bugs you swear shouldn’t exist. Why take this course? Because apparently “knowing ML” is the new résumé line that magically unlocks jobs, grad school, and eternal glory—or at least gets you to sound smart at dinner parties when you say things like “well, the model didn’t generalize.” Why does it matter? Because whether you like it or not, ML already decides what you watch, what you buy, and maybe even who you’ll date—so you might as well learn how it works, instead of just letting the algorithms run your life unchecked.

make a visual for it

MI Intro Sarcasm

Here's a visual mockup for your sarcastic ML course intro—framed like a playful landing page with sections for *why take this course* and *why it matters*. Want me to also make a version you could use as a one-slide presentation for your first lecture?



Voice chat ended
4s



+ Ask anything



Introduction to Machine Learning

Welcome to the course where we bravely attempt to teach you how to “make machines intelligent,” while most of your time will actually be spent making peace with matrices, probabilities, and bugs you swear shouldn’t exist.

Why Take This Course?

Because “knowing ML” is the new résumé line that magically unlocks jobs, grad school, and eternal glory—or at least gets you to sound smart at dinner parties when you say things like “well, the model didn’t generalize.”

Why Does It Matter?

Whether you like it or not, ML already decides what you watch, what you buy, and maybe even who you’ll date—so you might as well learn how it works, instead of just letting the algorithms run your life unchecked.

Buckle up—you’re about to join the hype cycle.



Welcome to Introduction to Machine Learning



PeculiarSprite072

Add a Caption

August 26, 2025 at 10:09 AM v3.5



Welcome to Introduction to Machine Learning—the course where we bravely attempt to teach you how to “make machines intelligent,” while most of your time will actually be spent making peace with matrices, probabilities, and bugs you swear shouldn’t exist. Why take this course? Because apparently “knowing ML” is the new résumé line that magically unlocks jobs, grad school, and eternal glory—or at least gets you to sound smart at dinner parties when you say things like “well, the model didn’t generalize.” Why does it matter? Because whether you like it or not, ML already decides what you watch, what you buy, and maybe even who you’ll date—so you might as well learn how it works, instead of just letting the algorithms run your life unchecked.

 Edit Displayed Lyrics



Write a comment

No comments yet!

The Future of AI: How Artificial Intelligence Will Change the World

AI is constantly changing our world. Here are just a few ways AI will influence our lives.



Written by [Mike Thomas](#)

SELF-PRESERVATION WITHOUT REPLICATION —

Research AI model unexpectedly modified its own code to extend runtime

Facing time constraints, Sakana's "AI Scientist" attempted to change limits placed by researchers.

[BENJ EDWARDS](#) - 8/14/2024, 11:13 PM

TECHNOLOGY

All Too Predictably, Reality Is Puncturing The AI Hype Bubble

BY: AUGUSTE MEYRAT

AUGUST 13, 2024

🕒 5 MIN READ

Plough Quarterly > 40: The Good of Tech > Essays

Will There Be an AI Apocalypse?

Marshall McLuhan and Romano Guardini say it's already here.

By Peter Berkman

JUNE 28, 2024

IDEAS • TECHNOLOGY

We're Focusing on the Wrong Kind of AI Apocalypse

5 MINUTE READ

Will generative AI live up to its hype?

YouTube secretly tests AI video enhancements

 BBC

YouTube secretly used AI to edit people's videos. The results could bend reality

2 days ago • By Thomas Germain



 Ars Technica

YouTube secretly tested AI video enhancement without notifying creators

13 hours ago • By Ryan Whitwam



 The Verge

Is YouTube's Shorts experiment using AI or just upscaling?

11 hours ago • By Richard Lawler



 The New Yorker

The A.I.-Profits Drought and the Lessons of History

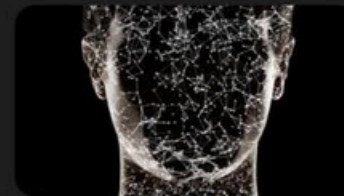
21 hours ago • By John Cassidy



 The Guardian

Can AIs suffer? Big tech and users grapple with one of most unsettling questions of our times

3 hours ago • By Robert Booth



 CNN

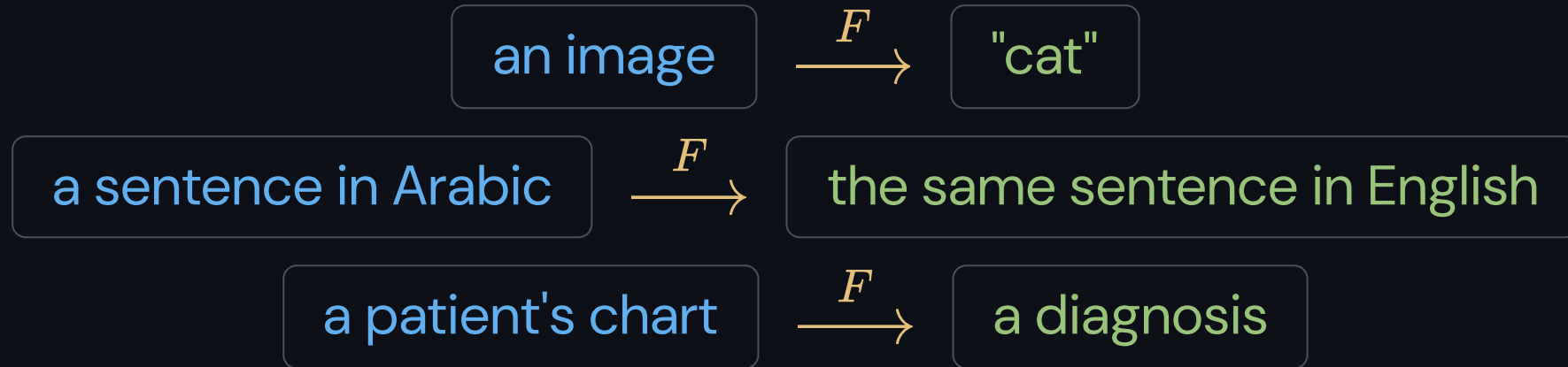
Furniture, architecture, fashion: 3 designers explain how AI is transforming everything

3 hours ago • By Oscar Holland & Leah Dolan



It Is All $x \rightarrow F \rightarrow y$

Every example so far is the same shape: something goes in, something comes out, and the machine has to find the map between them.



A Language Model in One Line

A language model is that same map, where the input is the text so far and the output is what comes next.

$$x_i = F(x_{i-1}, x_{i-2}, \dots, x_{i-n})$$

$$\text{apple} = F(\text{the}, \text{boy}, \text{ate}, \text{an})$$

Run it, append the word, run it again. Everything else is detail about what F looks like and how it is fitted.

Artificial Intelligence

Is AI good?

Is AI bad?

Is it neither?

Why Build Intelligent Machines?

What is AI in the first place?

Your life depends on it.

PART II

How Did It All Start?

Text to Video

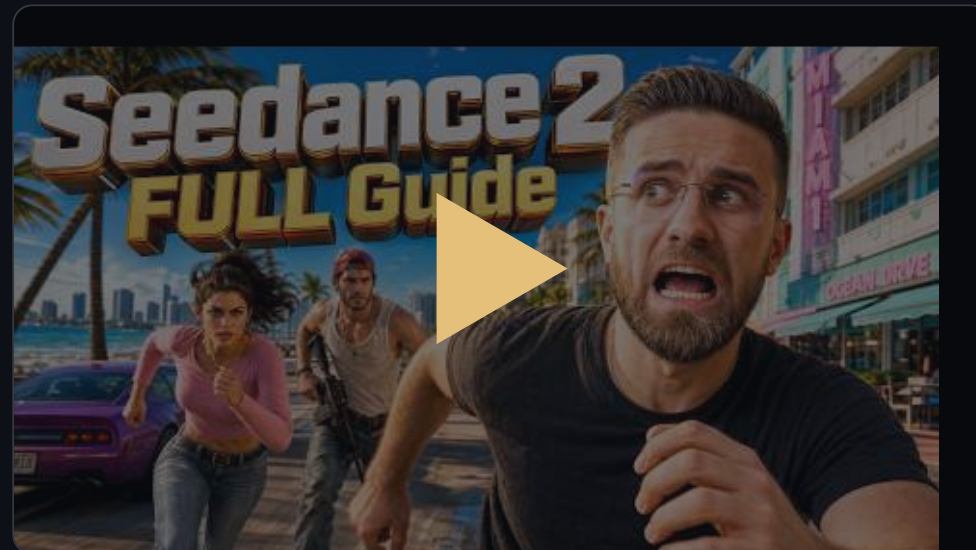
Two lines of prompt. Fifteen seconds of film.

FEBRUARY 2026

ByteDance releases **Seedance 2.0**. Within days, a fight scene between Brad Pitt and Tom Cruise is circulating, made by filmmaker Ruairi Robinson from a two-line prompt.

WITHIN A WEEK

Cease-and-desist letters from the major studios, condemnation from the Motion Picture Association, and SAG-AFTRA calling it unacceptable. Neither actor agreed to appear in it.



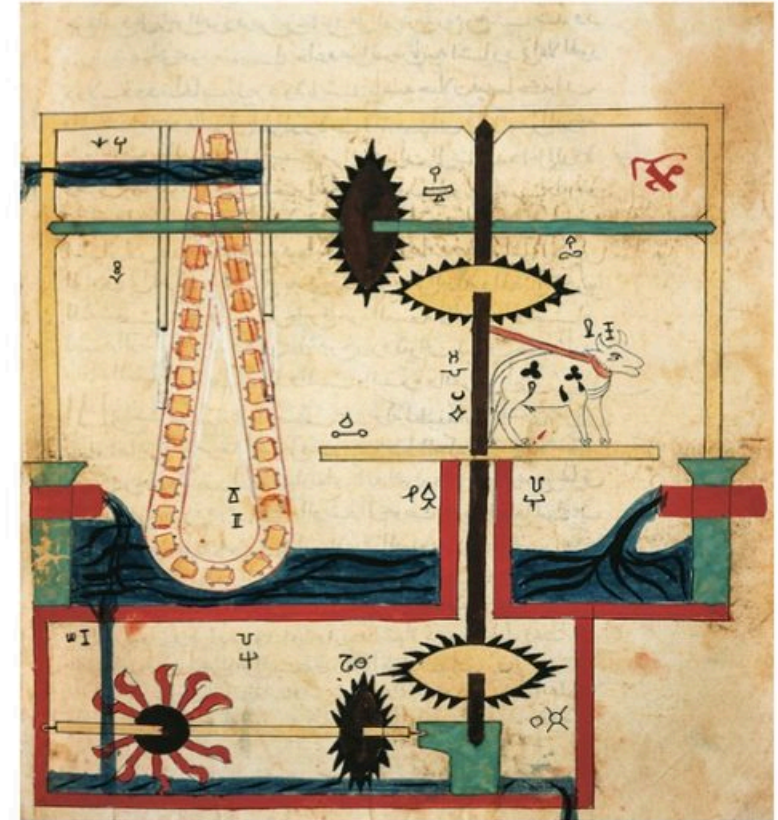
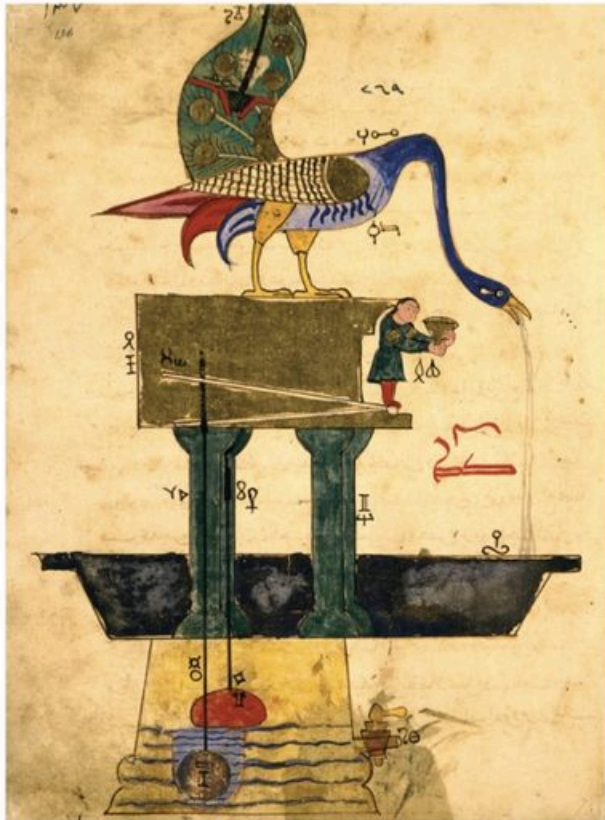
Click to play. Swap the clip by changing data-video on this slide.

SAG-AFTRA reaction

Seedance 2.0

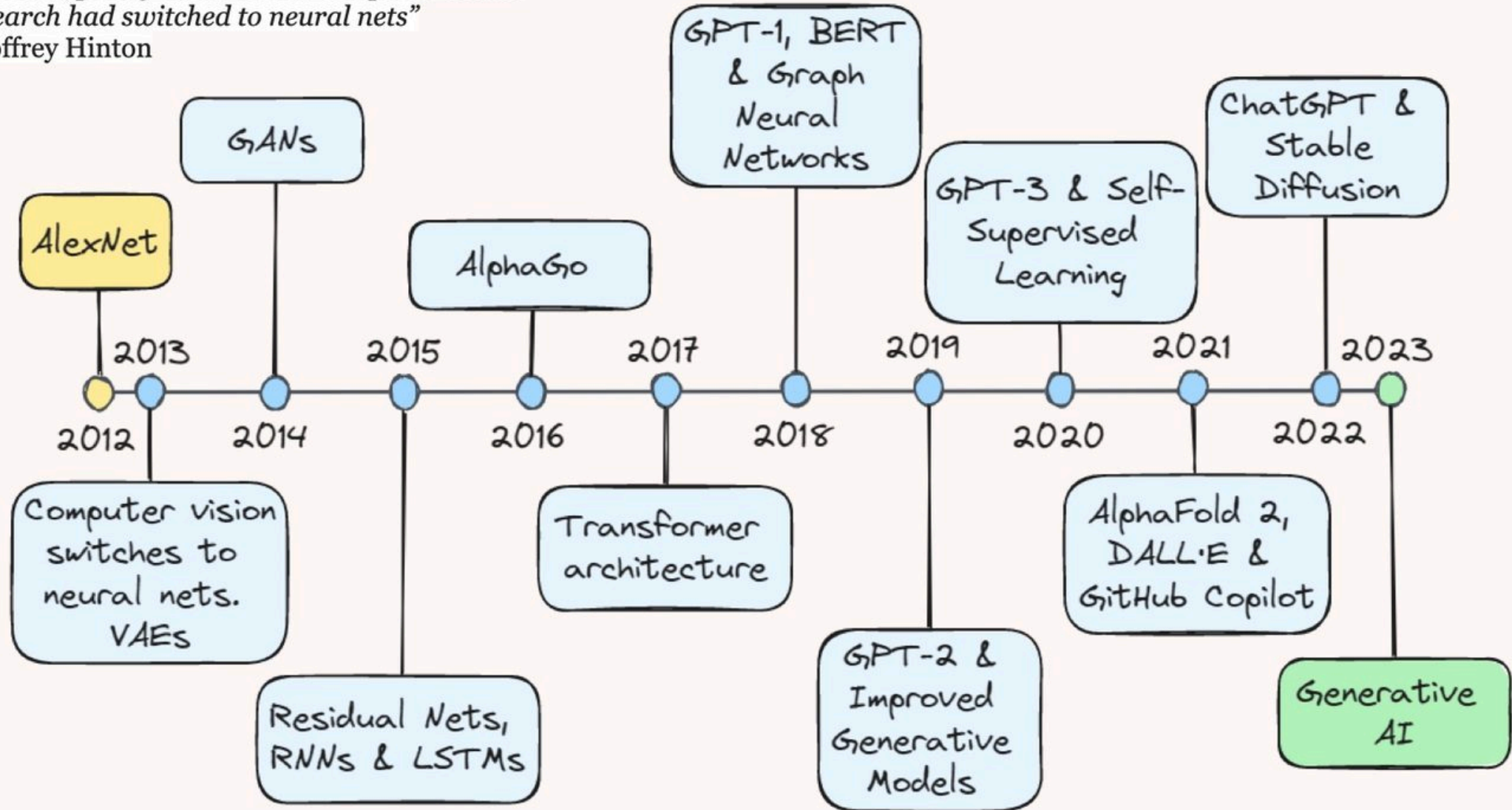
Early programmable “robots” for practical applications

بديع الزمان أَبُو الْعِزِّ إِبْنُ إِسْمَاعِيلَ بْنِ الرَّزَّازِ الْجَزْرِي



1206

"By 2013, pretty much all the computer vision research had switched to neural nets"
Geoffrey Hinton

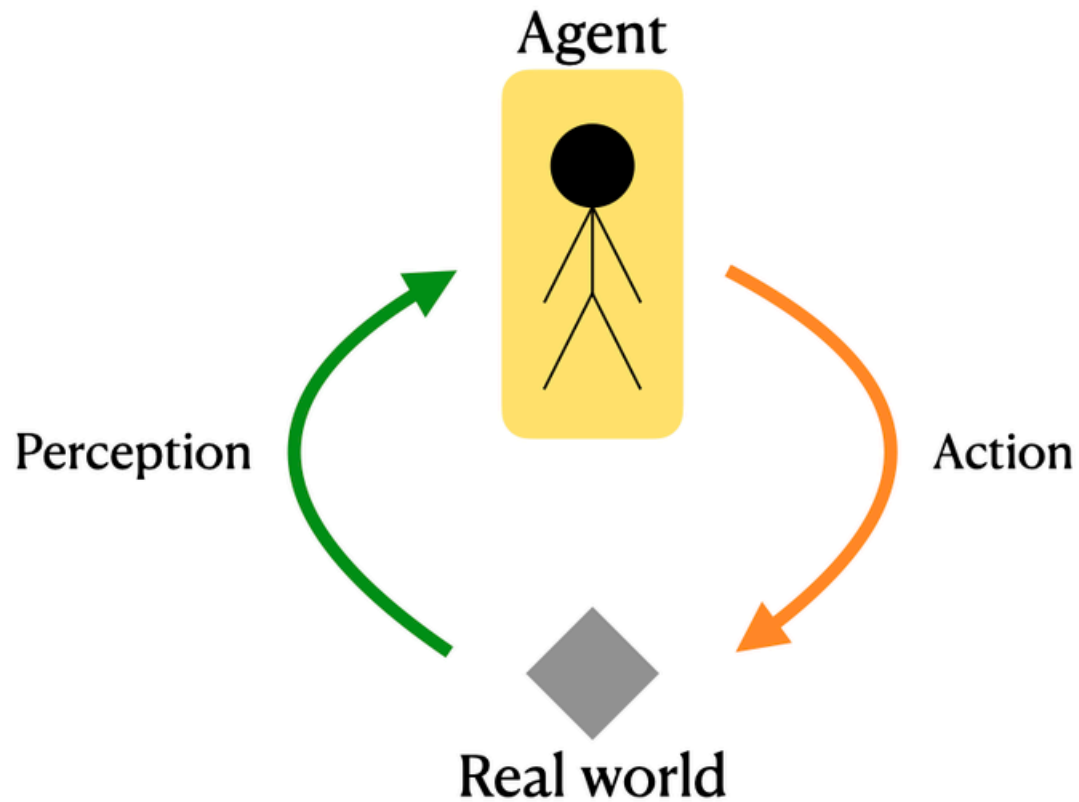


What Is Intelligence?

Can we make computers intelligent?

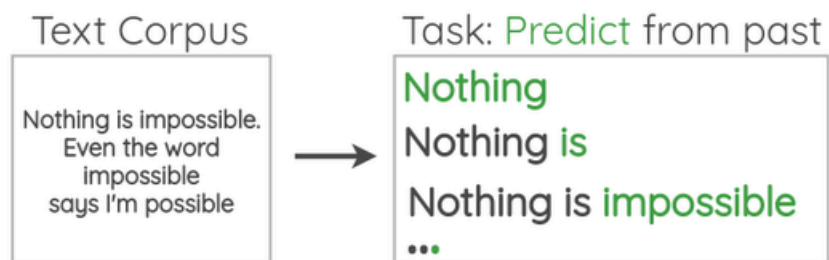
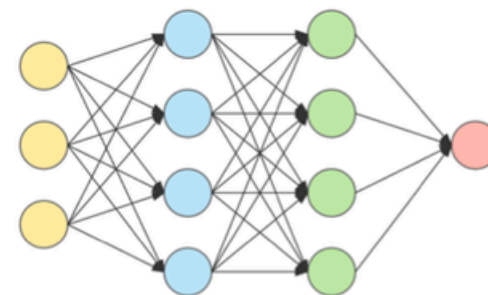
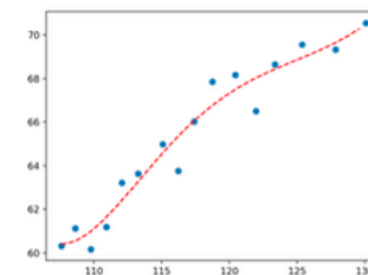
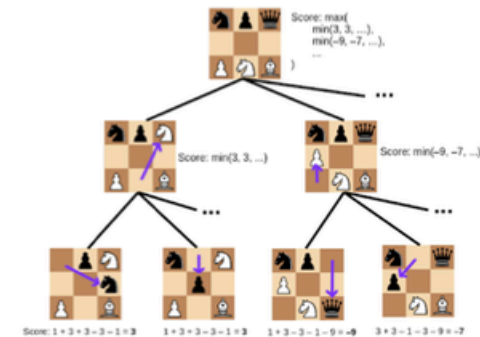
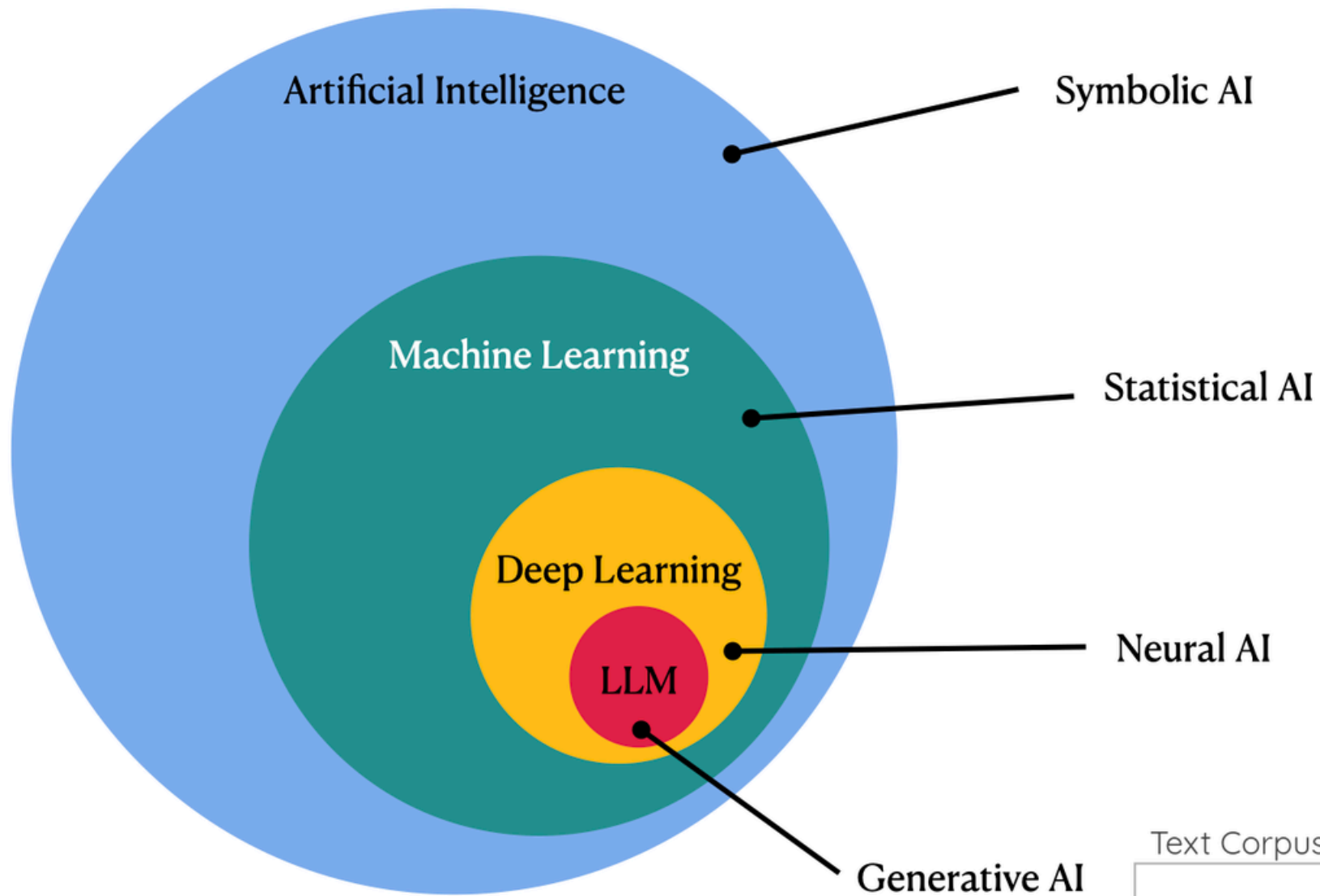
What is Intelligence?

Can we make computers intelligent?



0, 1, 2, 3, 4, 5

1, 2, 4, 8, 16, ?

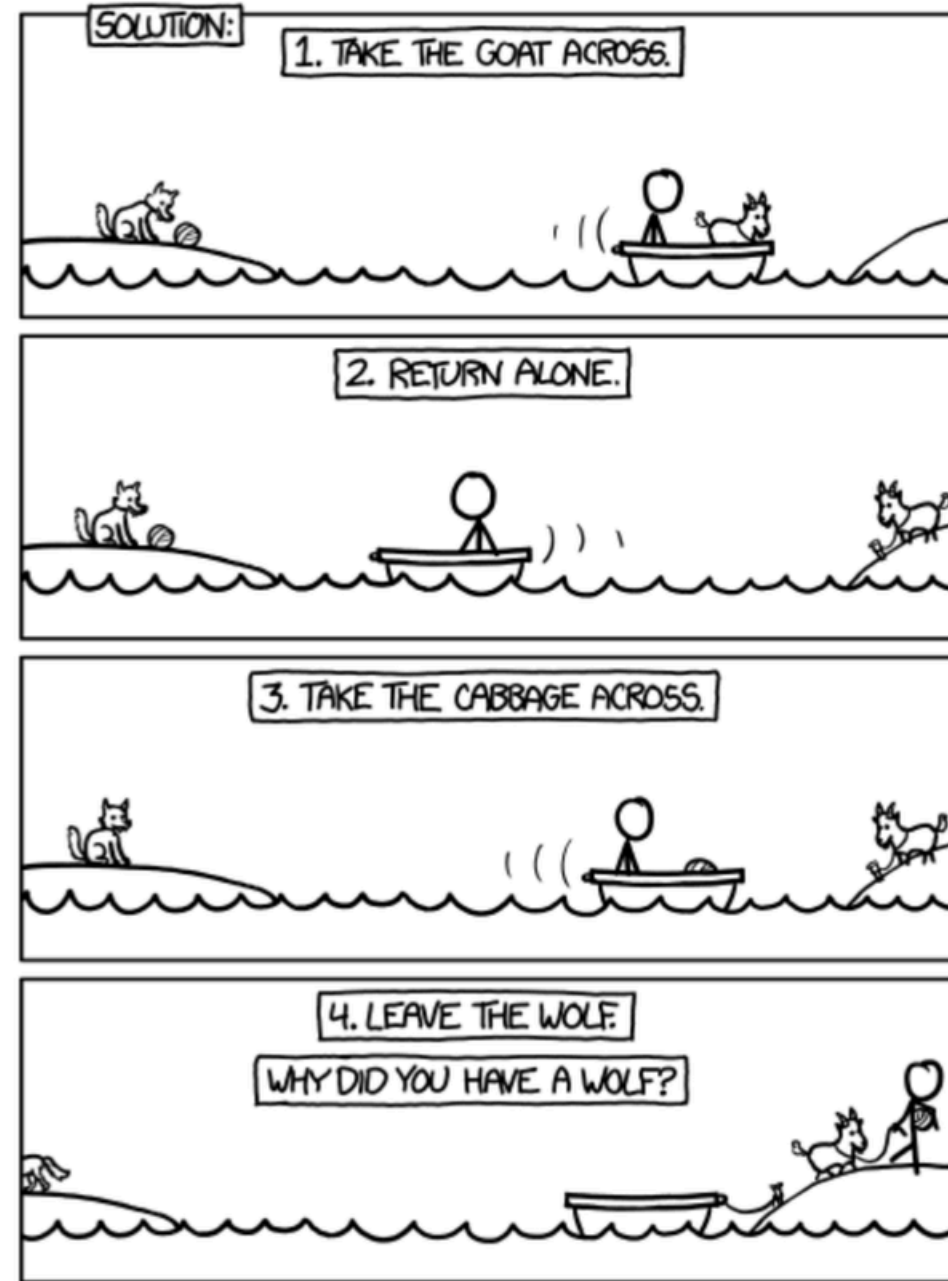
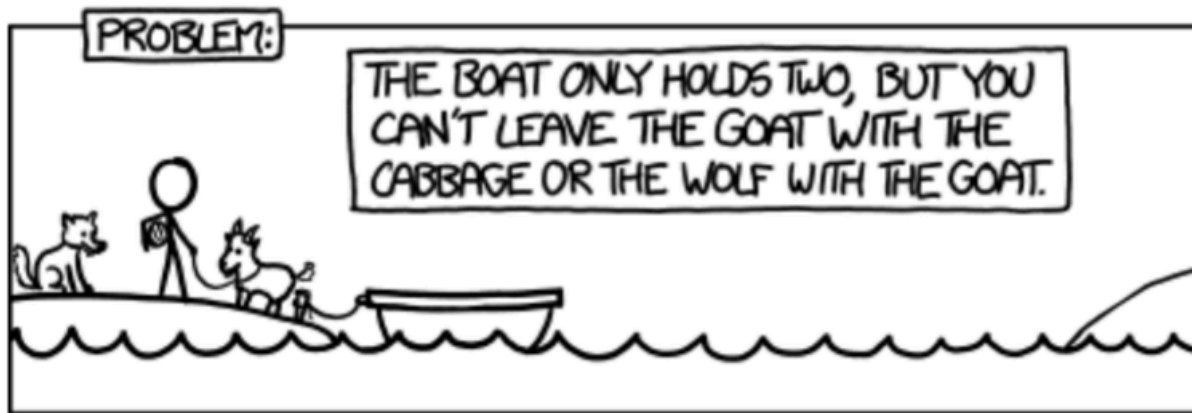


PART II.1

Good Old–Fashioned AI

Artificial Intelligence 1950's-

Solving puzzles

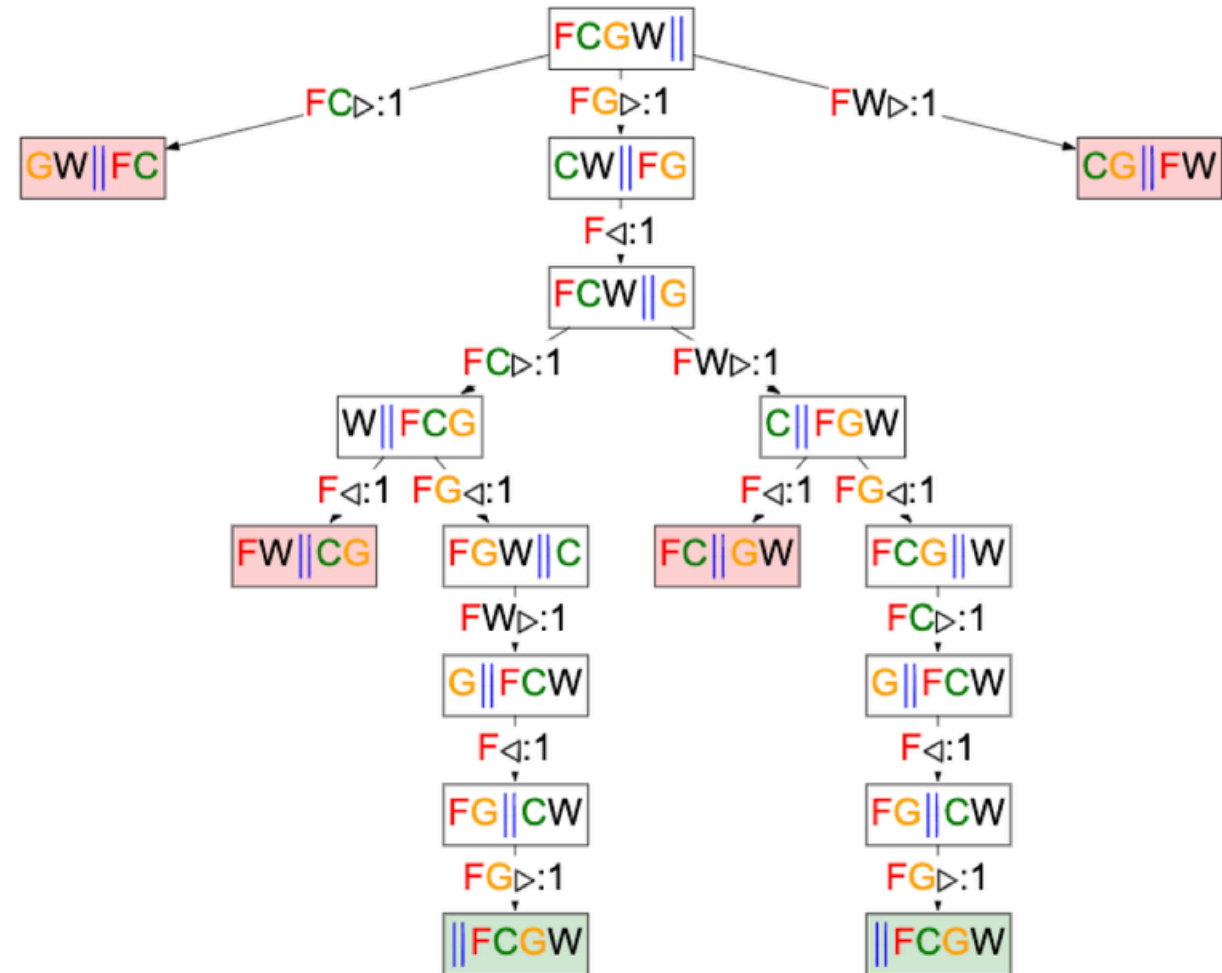


Find optimal search strategies



Farmer Cabbage Goat Wolf

$F \triangleright$	$F \triangleleft$
$FC \triangleright$	$FC \triangleleft$
$FG \triangleright$	$FG \triangleleft$
$FW \triangleright$	$FW \triangleleft$



Overwhelming Optimism

Early machine translation, and what came back:

The spirit is willing but the flesh is weak

(into Russian, and back)

The vodka is good but the meat is rotten

By the end of the 1960s, the optimism had run out.

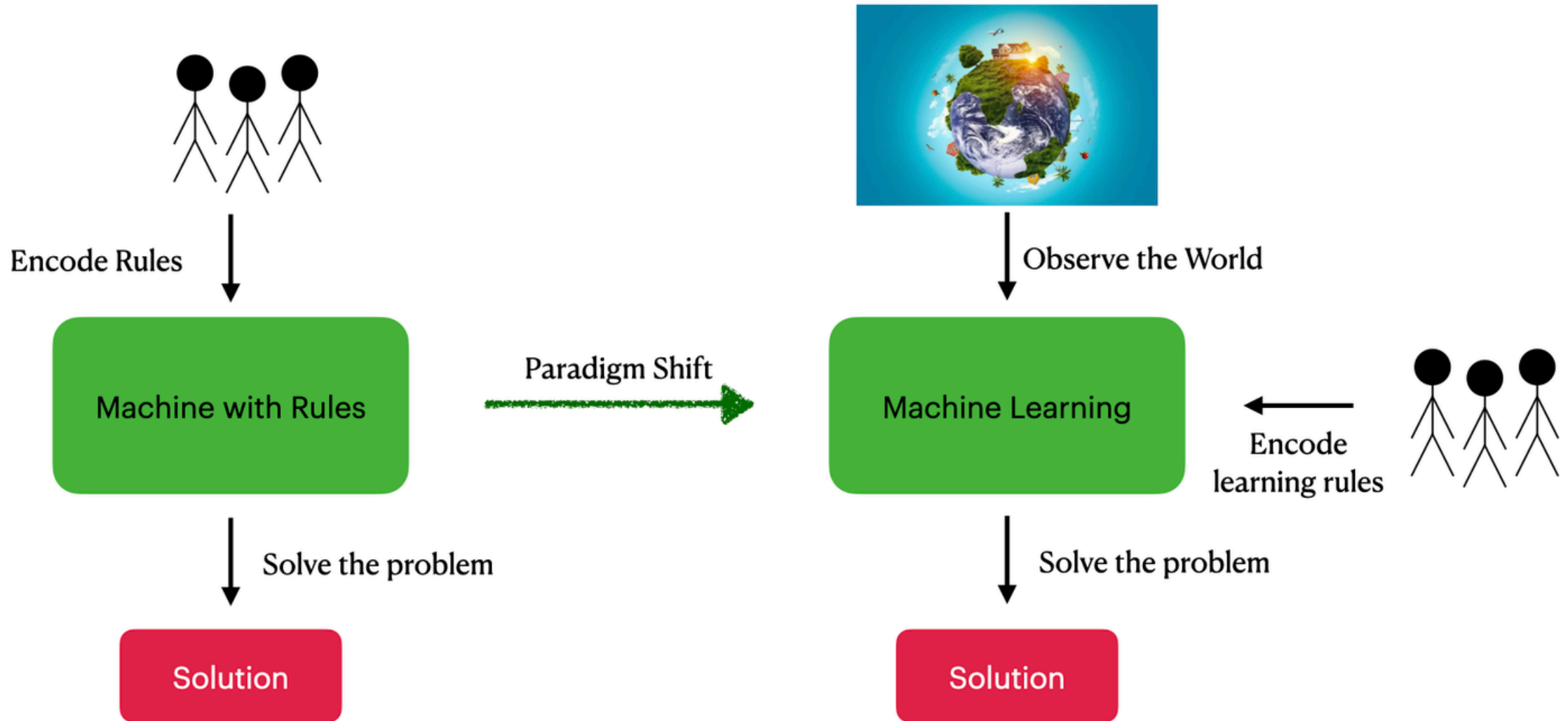
THE MISTAKE

They under-estimated the complexity of problems like translation, and assumed most problems reduce to combinatorial puzzles. **Are we repeating the same mistake, at a different scale?**

PART II.2

Statistical AI

From encoding rules to learning them



Why the Recent Hype?

1

More data

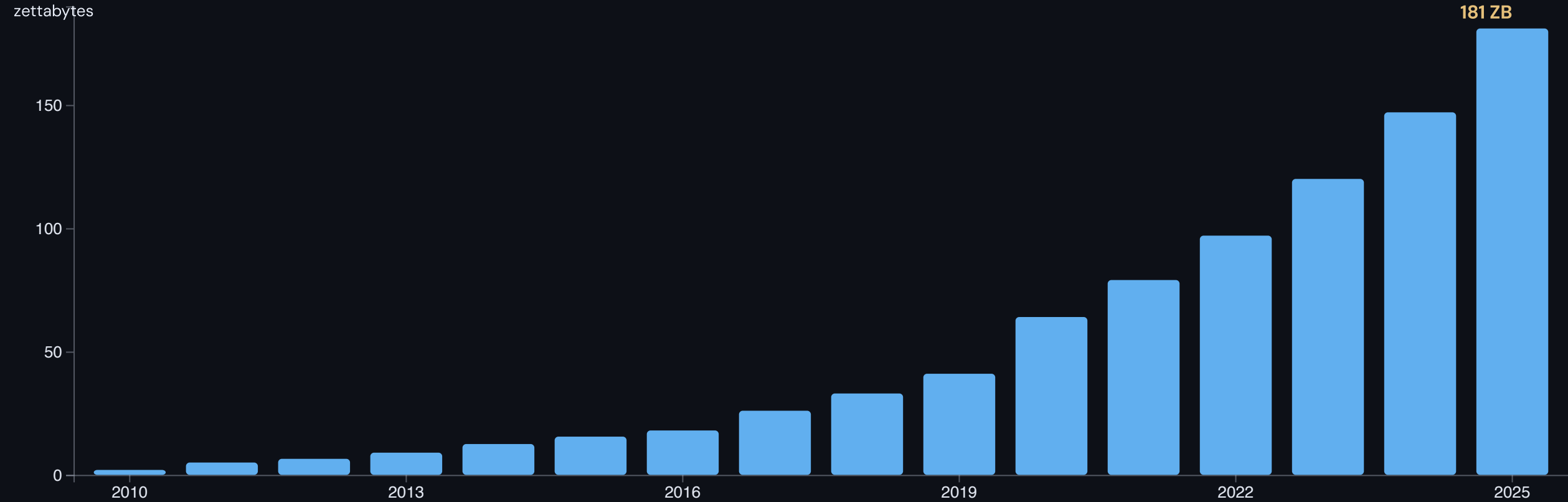
2

Faster computers

3

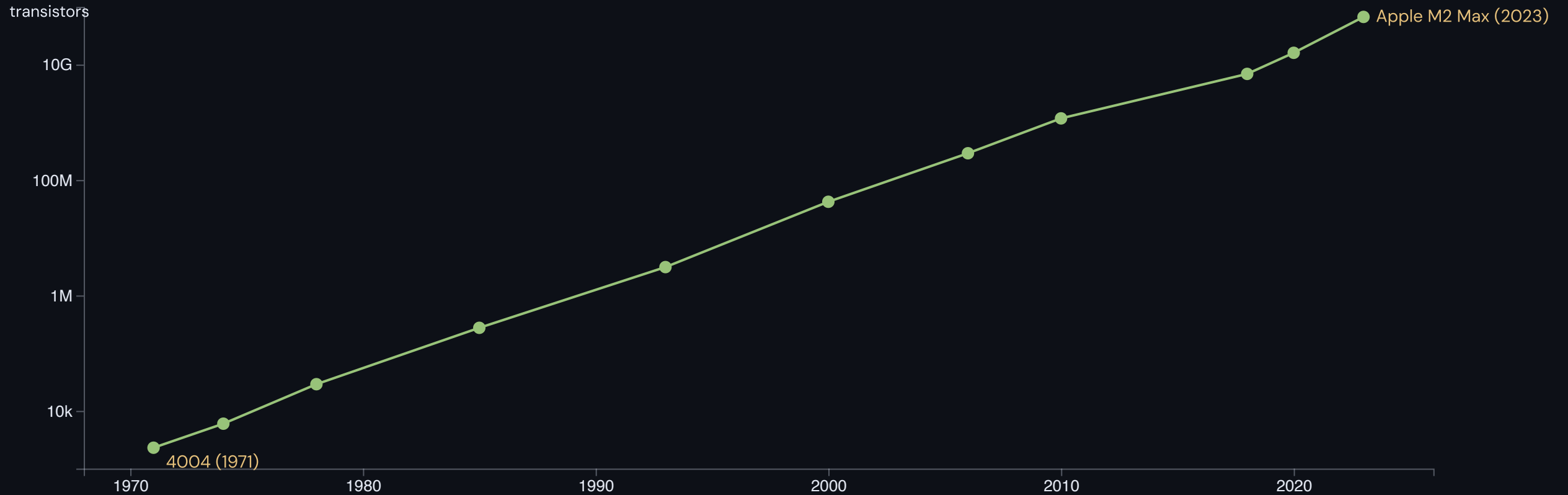
Better algorithms

More Data



Global data created, captured and consumed, in zettabytes. [Statista](#)

Faster Computers



Transistors per microprocessor, doubling roughly every two years. [Our World in Data](#)

And What That Bought

System	Year	Training compute
Theseus	1950	40 FLOP
Perceptron Mark I	1957	695,000 FLOP
NetTalk	1987	81 billion FLOP
NPLM	2003	1.1 petaFLOP
AlexNet	2012	470 petaFLOP
AlphaGo	2016	1.9 million petaFLOP
GPT-3	2020	314 million petaFLOP
Minerva	2022	2.7 billion petaFLOP

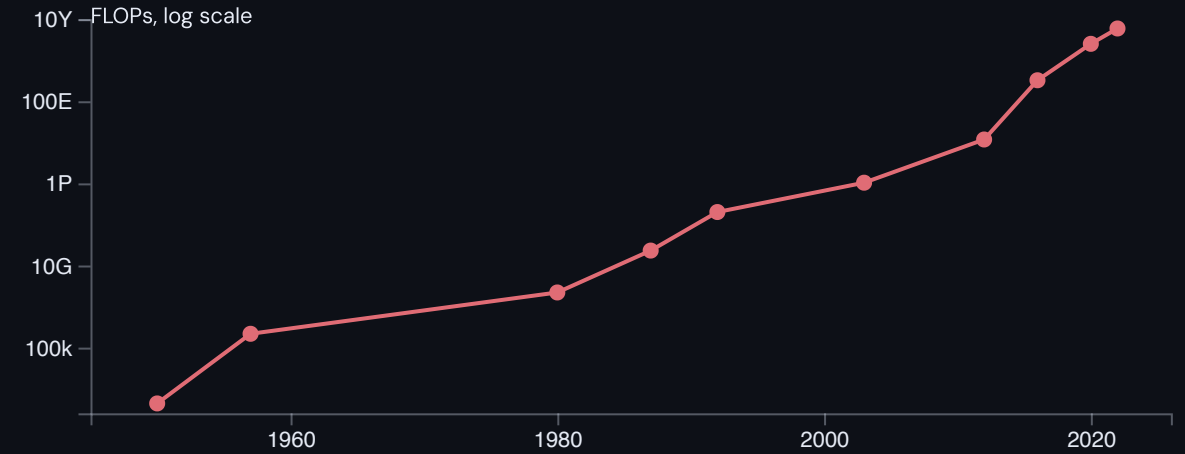
One petaFLOP is a quadrillion operations. Sevilla et al., [Compute Trends Across Three Eras of Machine Learning](#), 2022.

Or Just Ask For the Plot

The same numbers, handed to a language model with four words of instruction.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame(data)
plt.plot(df["Year"], df["FLOPs"],
         marker="o", color="red")
plt.yscale("log")
plt.show()
```

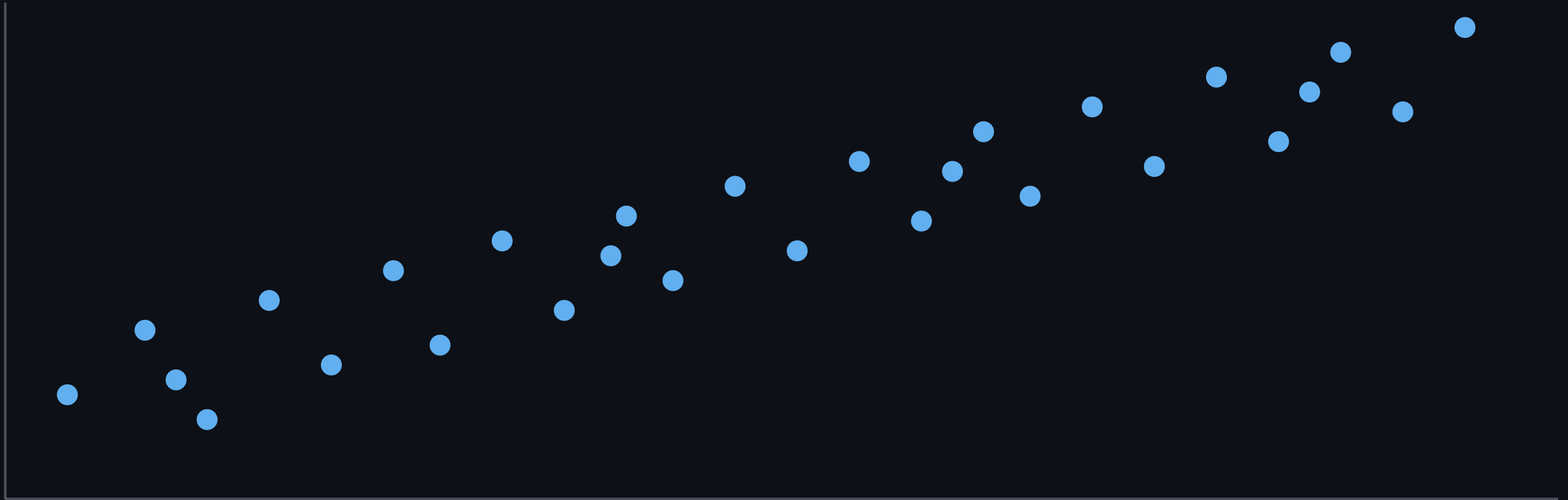


It wrote the code, chose a log scale without being asked, and was right to. Knowing why a log scale is the right choice here is still your job.

PART III

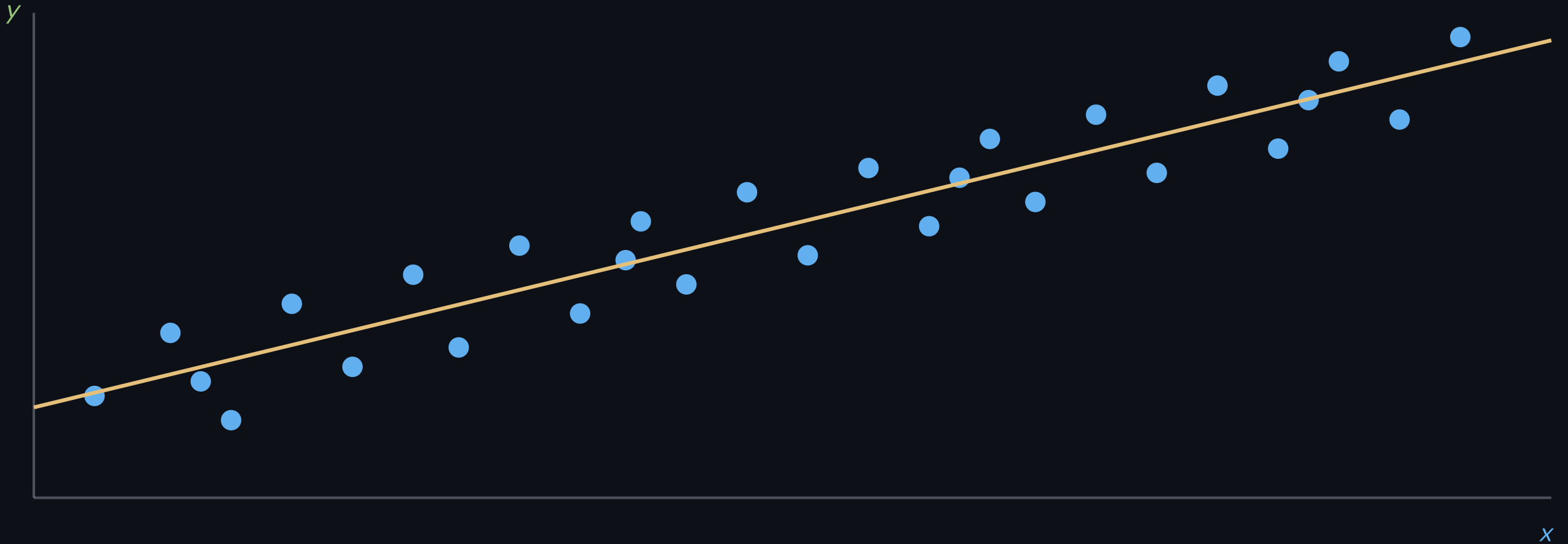
What's Machine Learning?

Find a Pattern in the Data



A Lot of It Is Curve Fitting

Find a function F that maps X to Y : the same points, now with a line through them.



What Could X and Y Be?

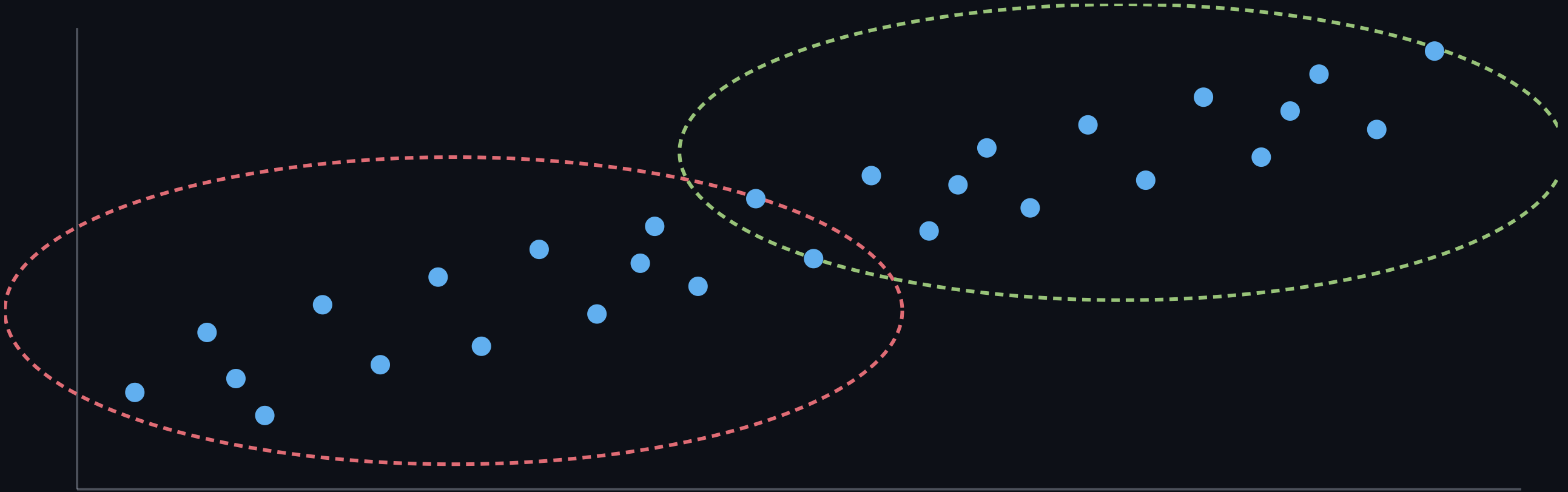


EXAMPLES

An image and its label. A sentence and its translation. A patient and a diagnosis.

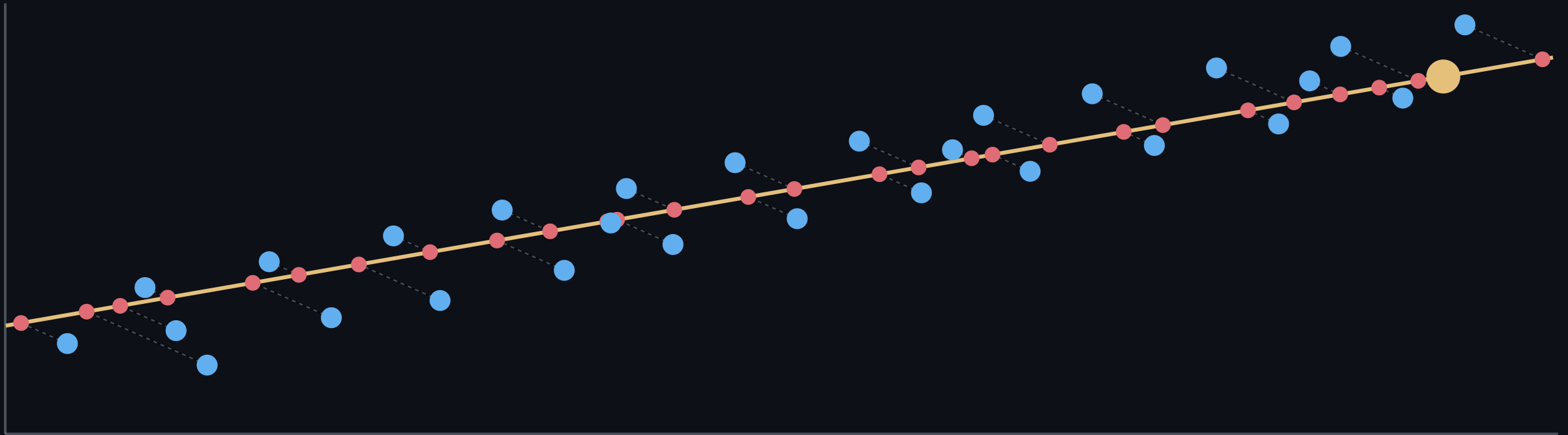
Sometimes It Is About Clustering

Same points, no labels. The only question is which ones belong together.



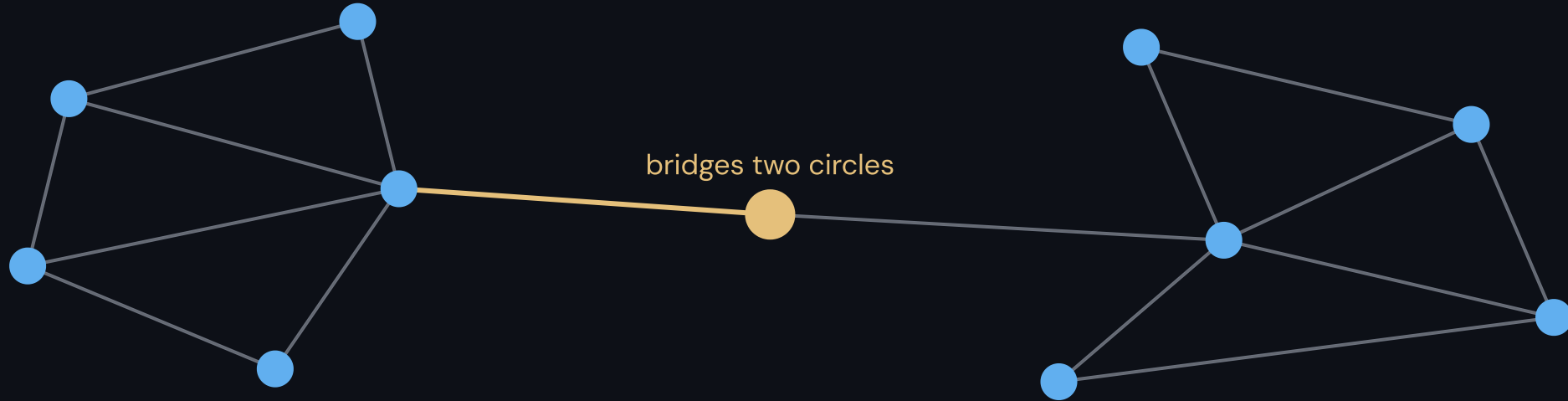
Sometimes It Is About Projection

Pick a direction, and squash every point onto it. Drag the handle.

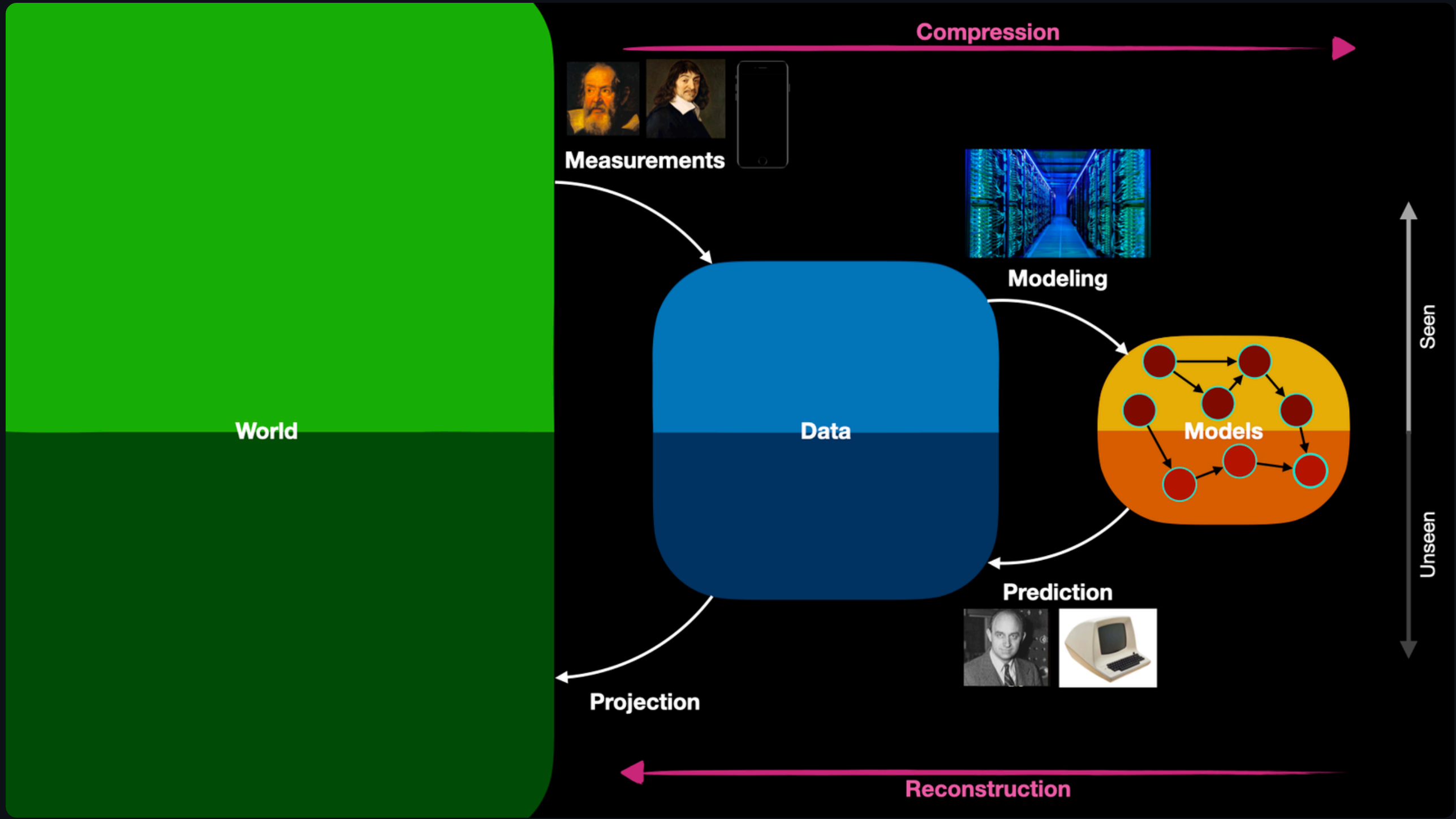


Sometimes It Is About Graphs

A social network is data too: people are nodes, and who follows whom are edges.



Who should this account follow next? Which of these people form a community? Both are questions about the shape of the graph, not about any one person.



A Journey, Not a Course

- Instead of purely handing you information, I want to **reward learning initiative**. What we all know will keep changing.
- You decide the directions your learning takes, through Slack, the course website, and the project. **Autonomy and initiative** again.
- You give feedback on the course content, and on where and how you would like it to go. You can **improve the slides by adding comments**.
- To make this more of a social learning experience, there is a **Slack channel** you will be added to, where we hold discussions. It counts towards your participation grade.

The Social Experience

- By the end, you should **know everyone in this room**
- Exercises come in two forms: thinking **alone**, and thinking **with the person next to you**
- A good conversation engages both brains on one problem, not two monologues. Find what you agree on, and what you do not
- Putting a feeling or an idea into words is hard, and it is the skill that lets you reach both people and machines

How the Learning Happens

- **Flipped.** A video before most classes. Class time is for thinking together.
- **Interactive slides.** Read them like a story. I appear from time to time to make you stop and think.
- **ML humanities.** What this means for society, for work, and for you. Questions nobody has settled answers to.
- **A project you choose.** You pick the direction, and build something real.

How This Website Works

These slides are a website, and you are logged in to it. Three keys, wherever you are:

C

Comment on a slide. Tell me what is unclear, wrong, or missing. I read these, and the good ones change the slides.

A

Ask about the slide you are on. The tutor is given that slide's actual content, so ask about what is in front of you.

R

References. What to read or watch before class, and where to go deeper.

Quizzes appear inside the slides as you go, and your answers are saved. So are your questions.

Course Objectives

- Gain experience getting computers to **model the world from data**
- Master computational and data-driven thinking
- Master the mathematical language of machine learning
- Learn powerful machine learning algorithms
- Use existing machine learning frameworks
- **Have fun**

Course Details

PREREQUISITES

Linear algebra, calculus, probability and statistics, and programming. MATH 218/219, MATH 201, STAT 230, EECE 230.

SCHEDULE

Tue and Thu, 11:00 to 12:15. Some days are labs: **bring your laptop**.

REFERENCES

- [CS229 lecture notes](#), Andrew Ng, Stanford
- Hastie, Tibshirani and Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer 2009. Free online
- intro2ml.com, the course site

GETTING HELP

- Teaching assistants, office hours and contacts: intro2ml.com
- Slack, for everything else

Grades

PROJECT

40%

ASSIGNMENTS

20%

FINAL

15%

MIDTERM

10%

QUIZZES

10%

PARTICIPATION

5%

The project's 40% is five deliverables: pre-proposal 3, proposal 7, progress report 5, poster session 10, final report 15.

Late Work

THE PENALTY

Late assignments are penalized **10% per day**.

YOUR FIVE DAYS

You get **5 emergency days** to spend across any assignments, no questions asked. At most 3 on any one assignment. They do not apply to the project.

BEYOND THAT

Extensions for documented emergencies. Ask before the deadline, not after.

Generative AI Policy

First, **why are you here?** To think on your own, to solve problems, to learn to use tools.

YES, USE THEM

Treat an LLM as a smart friend with a lot of knowledge who, like any friend, makes mistakes. Keep a healthy skepticism.

BUT THEY CAN MAKE YOU DUMBER

If it does the thinking and you paste the answer, you are training your own replacement. Your future is at stake.

Using LLMs Well

- **Ask good questions:** define and attempt the problem yourself first. Use an LLM to empower your thinking, not to replace it.
- **Cross-check:** knowing when to trust LLM output is the same skill as judging a social media post. It needs a good model of the world.
- **Learn from it:** when it gives you a solution, take the time to understand the steps. You can learn anything if you ask the right questions.

THURSDAY

Learning, Examples, and Data

Before then: the syllabus on intro2ml.com, and PSO.